Bit - manipulation

Operators

① AND

a	b	a & b
0	0	0
0	1	0
1	0	0
1	1	1

★ when you & 1 with any number, digits remain the same.

$$\begin{array}{r} 1\,1\,0\,0\,1\,0\,1\,0\,0 \\ \&\ 1\,1\,1\,1\,1\,1\,1\,1\,1 \\ \hline 1\,1\,0\,0\,1\,0\,1\,0\,0 \end{array}$$

② OR

a	b	a OR b
0	0	0
0	1	1
1	0	1
1	1	1

③ XOR (^) (if and only if)

exclusive OR

a	b	a ^ b
0	0	0
0	1	1
1	0	1
1	1	0

observations

$a \,\hat{}\, 1 = \bar{a}$

$a \,\hat{}\, 0 = a$

$a \,\hat{}\, a = 0$

④ Complement ($\sim$)

$a = 1\ 0\ 1\ 1\ 0$

$\bar{a} = 0\ 1\ 0\ 0\ 1$

Continued ahead

Number Systems

① Decimal $\rightarrow$ 0, 1, 2, . . . , 9 Base : 10

$(357)_{10}$, $(10)_{10}$

② Binary $\rightarrow$ 0 & 1 Base : 2

$(10)_{10} = (1010)_2$

$(7)_{10} = (111)_2$

③ Octal $\rightarrow$ 0, 1, 2, 3, . . . 7 Base : 8

Decimal: 0, 1, 2, 3, 4, 5, 6, 7 → 8, 9, 10, 11, 12, 13, 14, 15

Octal: 0, 1, 2, 3, 4, 5, 6, 7, 10, 11, 12, 13, 14, 15, 16,
17, 20

$$(9)_{10} = (11)_8$$

4 Hexadecimal: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

Base: 16

$$(10)_{10} = (A)_{16} \qquad (12)_{10} = (C)_{16}$$

Conversations:

① Decimal to base b

Q: Convert $(17)_{10}$ to base 2

Keep dividing by base, take remainders, write in opposite.

$$
\begin{array}{r|l}
2 & 17 \\ \hline
2 & 8 \;-\; 1 \\ \hline
2 & 4 \;-\; 0 \\ \hline
2 & 2 \;-\; 0 \\ \hline
 & 1 \;-\; 0
\end{array}
$$

$(10001)_2 = (17)_{10}$

a

1	0	0	0	1

$(17)_{10} = (?)_8 \Rightarrow (21)_8$

$$8 \,\underline{|\; 17 \;}\; 1$$
$$2 - 1$$

② Convert any base b to decimal.

$(10001)_2 = (\quad)_{10}$?

Steps: multiply & add the power of base with digits.

$$= 1 * 2^4 + 0 * 2^3 + 0 * 2^2 + 0 * 2^1 + 1 * 2^0$$

$$= 16 + 0 + 0 + 0 + 1 = (17)_{10}$$

Q: $(21)_8 = (\quad)_{10}$

$$= 2 * 8^1 + 1 * 8^0$$

$$= 2 \times 8 + 1 = (17)_{10} \checkmark$$

Continuing with operators:

5 Left shift operator $(<<)$

$(10)_{10} = (1010)_2$ $\qquad$ $10 << 1$

Step! $\qquad 1010 << 1 = 10100$

$$1 * 2^4 + 0 * 2^3 + 1 * 2^2 + 0 * 2^1 + 0 * 2^0 = 16 + 4 = 20$$

$$\therefore\ a << 1 = 2a$$

General point:

$$a << b = a * 2^b$$

⑥ Right shift: >>

Q:

$001100\underline{1} >> 1 \Rightarrow 001100$

$$(00,11234)_{10} = (11234)_{10}$$

$$(1100)$$

(Ignored) → Same for all number systems.

General point:

$$\boxed{a >> b = \frac{a}{2^b}}$$

Questions:

① Given a no. n find if it is odd or even.

Point: Every no. is calculated in binary form internally.

$12 + 7 \Longrightarrow$

$$\begin{array}{r} 1100 \\ +\ 0111 \\ \hline 10011 \end{array}$$

$(19)_{10} = (10011)_2$

$16 + 2 + 1$ ✓

Note:

$\boxed{1001}1$ — leaving this, every other is a power of 2

$\to 2^0$

This will always be even.

Hence: if 2^0 place $== 1 \Longrightarrow$ odd

otherwise $\Longrightarrow$ even

0 1 1 1 0 1 1 0 1 1 0 1 → LSB : Least Significant Bit.

$\underbrace{2^{n} + \dots 2^{1}}_{\text{Always is even}} + 2^{0}$

1 → odd

0 → even.

$$\begin{array}{r} 1\,0\,0\,1\,0\,1 \\ \&\; 0\,0\,0\,0\,0\,1 \\ \hline 1\;\; 0\,0\,0\,0\,0\,0\,1 \\ \hline \end{array} \Rightarrow (1) \text{ then, odd.}$$

Sum up: n & 1 == 1. ⇒ odd else even.

② arr = [2, 3, 4, 1, 2, 1, 3, 6, 4]

Ans

$$(5 \times 3) \times (5 \times 4) = 5 \times 5 \times 3 \times 4$$

$$= 5 \times 4 \times 3 \times 5$$

$$= 4 \times 3 \times 5 \times 5$$

$$2 \wedge 3 \wedge 3 \wedge 4 \Rightarrow 2 \wedge 4 \wedge 3 \wedge 3$$

Ans: XOR all the nos.

Q: [-2, 3, 2, 4, -5, 5, -4]

Q: Find i^{th} bit of a no.

8	7	6	5	4	3	2	1
1	0	1	1	0	1	1	0

?

Ans:

5^{th}

1 0 1 1 0 1 1 0

& 0 0 0 1 0 0 0 0

0 0 0 1 0 0 0 0 Ans

This is called ? a mask.

$n \Rightarrow$ mask with $n-1$ zeros $\rightarrow 1 << (n-1)$

$1 << 4 \Rightarrow 10000$

Ans: $n \,\&\, (1 << (n-1))$

Q: Set the i^{th} bit.

→ turn it to 1 → $0 \rightarrow 1$, $1 \rightarrow 1$

$$
\begin{array}{rcccccccc}
 & & & & 4^{th} & & & \\
 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\
\text{OR} & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
\hline
 & 1 & 0 & 1 & 1 & 1 & 1 & 0
\end{array}
$$

mask

Q: Reset i^{th} bit:

$1 \rightarrow 0$, $0 \rightarrow 0$

$$
\begin{array}{rcccccccc}
 & & & 5^{th} & & & & \\
 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\
\& & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\
\hline
 & 1 & 0 & 0 & 0 & 1 & 1 & 0
\end{array}
$$

Ans

How to get this mask?

$\underline{\text{mask}}$: $!(1 << (n-1))$

Q: Find the position of the right most set bit.

Ex: $\underbrace{101101}_{a}\,1\,\underbrace{00}_{b}$ Ans = 4

$N = a1b$

$-N = \bar{a}1b$

Ans = $N \,\&\, (-N)$

$a = 101101$

$b = 00$

How?

Negative of a number in Binary form:

1 byte = 8 bits

MSB

LSB

10 ⟹

0	0	0	0	1	0	1	0

→ value of no.

→ Tells us if no. is +ve or -ve

1 → -ve

0 → +ve

−10 = ?

Steps:

1. Complement of no.
2. +1 to it

⟹ aka 2's complement method.

$$(10)_{10} = (00001010)_2$$

① $1\ 1\ 1\ 1\ 0\ 1\ 0\ 1$

②

$$
\begin{array}{r}
 \quad\;\; ^{1} \\
1\ 1\ 1\ 1\ 0\ 1\ 0\ 1 \\
+ \qquad\qquad\qquad 1 \\
\hline
(1\ 1\ 1\ 1\ 0\ 1\ 1\ 0)_2
\end{array}
\Rightarrow (-10)_{10}
$$

why two?

$$1\ 0\ \boxed{\bar{1}\ \bar{0}\ \bar{1}\ \bar{1}\ \bar{0}\ \bar{1}\ \bar{1}\ \bar{1}}$$

discarded.

$$
\begin{array}{r}
100000000 \\
-\ 00001010 \\
\hline
\end{array}
$$

What is this?

$100000000 = 11111111 + 1$

Now: $11111111 + 1 - 00001010$

$$\Rightarrow \underbrace{11111111 - 00001010}_{\text{complement}} \underbrace{+1}_{\text{step 2}}$$

$1000 = 111 + 1$

$8 \quad 7 + 1$

$10000 = 1111 + 1$

$16 \quad 15 + 1$

$$
\begin{array}{r}
1111 \\
+\ 0001 \\
\hline
10000 \\
\hline
\end{array}
$$ ✓

$$
\begin{array}{r}
1\;1\;1\;1\;1\;1\;1\;1 \\
-\;0\;0\;0\;0\;1\;0\;1\;0 \\
\hline
1\;1\;1\;1\;0\;1\;0\;1
\end{array}
$$

complement

Range of numbers.

① **1 byte :**

sign of number

0,1	0,1	0,1	0,1	0,1	0,1	0,1	0,1

128

Total $= 2 * 2 * 2 * \ldots$ 8 times

$= 2^8 = 256$

actual no is stored in bits = n-1 → total bits

In 1 byte : 7 bits

Total I can make from 7 bits

$= 2^7 = \underline{128}$

-128 to 127 → 0 is here as well

0 0 0 0 0 0 0 0

1 ⇒ 1 1 1 1 1 1 1 1

②

1 1 1 1 1 1 1

1 1 1 1 1 1 1 1

\+ 1

1 0 0 0 0 0 0 0 0 ⟹ 9 bits

1 will be discarded.

$-0 = 0$

Range formula: for n bits

$$\boxed{-2^{n-1} \text{ to } 2^{n-1} - 1}$$

Q:

arr = [2, 2, 3, 2, 7, 7, 8, 7, 8, 8]

						1	0

1 0

1 1

1 0

1 1 1

1 1 1

1 0 0 0

3	3	6	3

If 3 was not in array.

1 1 1

1 0 0 0

1 0 0 0

1 1 1 1 1 1 3 1 3 7 4 % 3

0 0 1 1 = 3

Q: Find the n^{th} magic no.

		5^3	5^2	5^1		magic no.
1	=	0	0	1	$\longrightarrow$	5
2	=	0	1	0	$\longrightarrow$	25
3	=	0	1	1	$\longrightarrow$	30 (5+25)
4	=	1	0	0	$\longrightarrow$	125
5	=	1	0	1	$\longrightarrow$	130
$\vdots$			$\vdots$			$\vdots$

n

$n = 6 \Rightarrow 110$

loop

$n \& 1 \Rightarrow$ This will give me last digit in binary.

$n >> 1$

$0 * 5^1 + 1 * 5^2 + 1 * 5^3$

Q: find no. of digits in base b.

$(6)_{10} \Rightarrow 1$

$(6)_{10} = (110)_2 = 3$

Formula?

$\log_b a = x$

$a = b^x$

$\log_2 10 = 3.32$

$10 = 2^{3.32}$

int +1

$\log_2 6 = x$

$6 = 2^x$

= no. of digits.

Formula:

No. of digits in base b of no. n

$$= \text{int}\left(\log_b n\right) + 1$$

$$\boxed{\log_b a = \frac{\log_x a}{\log_x b}}$$

Pascal's Triangle :

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

1 5 10 10 5 1

∴ Find the sum of n^{th} row.

Ans: Sum of each row =

$$^nC_0 + {^nC_1} + {^nC_2} + \ldots + {^nC_n} = 2^n$$

For n^{th} row, sum $= 2^{n-1}$

Ans: 1 << (n-1) $= 1 \times 2^{n-1}$

Ans

Q: You are given a number. find out if its power of 2 or not.

100000 , 100010

$$10000000 = \underbrace{1111111}_{n-1} + 1$$

$$\begin{array}{r} 10000000 \\ \& \ 01111111 \\ \hline 0 \end{array} \qquad \begin{array}{r} 10010 \\ \& \ 01111 \\ \hline 00010 \end{array}$$

Ans: If $n \& (n-1) = 0$ // It is power of 2.

Q: Calculate a^b

// $O(b)$

$3^6 \Rightarrow 3 * 3 * 3 * 3 * 3 * 3$

$3^6 = 3^{110} = 3^{2+4} = 3^2 * 3^4$

ans = ~~1~~ 9

base = 3

base = 9

81

base = base * base

O(log(b))

n = 110

n & 1 ⟹ 0

n >> 1 ⟹ 11 & 1 ⟹ 1

n = 11 >> 1 ⟹ 1 & 1 = 1

n = 1 >> 1 = 0

110

3 ⟹ 4 2 0
$3 * 3 * 3 * 0$

ans =
ans * base

Imp to understand this question

Q: Given a number n, find the no. of set bits in it.

$n = 9$

$n = 1001$

Ans = 2

$n \& (-n) = 0001$

$n - [n \& (-n)] = 1000 \Rightarrow 1$

$$n = \quad 1001$$
$$\& \quad 1000$$

$$1000 \quad 1$$

$$8 \ \& \ 7 \Rightarrow \begin{array}{r} 1000 \\ \& \ 111 \\ \hline 0 \end{array} \quad 2$$

No. of set bits = no. of iterations.

Q: Find XOR of nos from 0 to a.

a	XOR from 0 to a
0	0
1	$0 \wedge 1 = 1$
2	$0 \wedge 1 \wedge 2 = 3$
3	0
4	4
5	1
6	7
7	0
8	8
9	1

If

	$0 \to a$
$a \% 4 = 0$	a
$a \% 4 = 1$	1
$a \% 4 = 2$	$a+1$
$a \% 4 = 3$	0

Q: XOR of all nos. between a & b

$a = 3$ & $b = 9$

$3 \wedge 4 \wedge 5 \wedge 6 \wedge 7 \wedge 8 \wedge 9$

$0 \wedge 1 \wedge 2 \wedge 3 \wedge 4 \wedge 5 \wedge 6 \wedge 7 \wedge 8 \wedge 9$

These are the extras

This is $0 \to (a-1)$

Ans $= \boxed{f(b) \wedge f(a-1)}$

$f(n) \to$ Xor of $0 \to x$

Q: Flipping an image:

$$\begin{matrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{matrix} \Longrightarrow \begin{matrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{matrix}$$

$$0 \wedge 1 = 1$$
$$1 \wedge 1 = 0$$

flipped

Ans

$$\Downarrow$$

$$\begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{matrix}$$